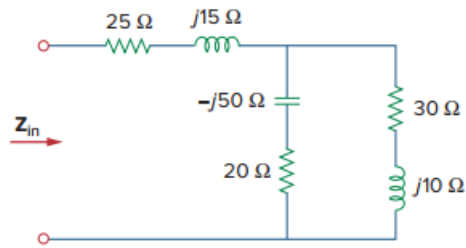


Problem

Obtain $\mathbf{Z}_{in}$ for the circuit in Fig. 9.67.

Figure 9.67:



Step-by-step solution

Step 1 of 5

Refer to Figure 9.67 in the textbook.

The $30\ \Omega$ resistor is in series with $j10\ \Omega$ inductor. Assume the equivalent impedance of their combination is represented by $\mathbf{Z}_1$. Therefore,

$$\mathbf{Z}_1 = 30 + j10\ \Omega$$

The $20\ \Omega$ resistor is in series with $-j50\ \Omega$ capacitor. Assume the equivalent impedance of their combination is represented by $\mathbf{Z}_2$. Therefore,

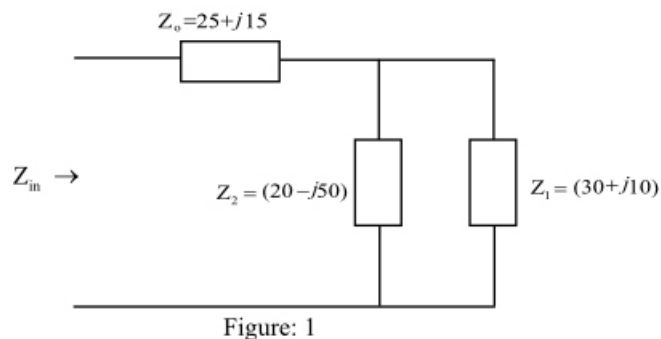
$$\mathbf{Z}_2 = 20 - j50\ \Omega$$

The $25\ \Omega$ resistor is in series with $j15\ \Omega$ inductor. Assume the equivalent impedance of their combination is represented by $\mathbf{Z}_0$. Therefore,

$$\mathbf{Z}_0 = 25 + j15\ \Omega$$

Step 2 of 5

Draw the figure containing the impedances $\mathbf{Z}_0$, $\mathbf{Z}_1$ and $\mathbf{Z}_2$ as shown in Figure 1.



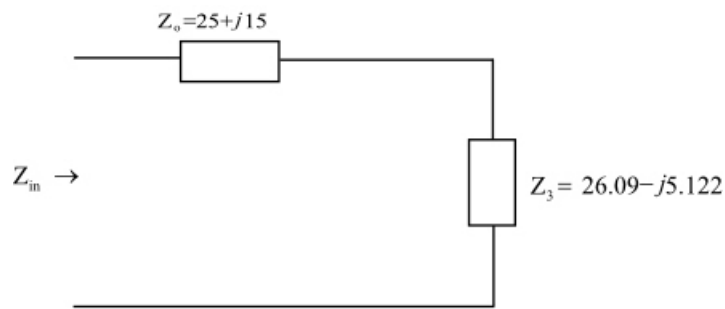
Step 3 of 5

In Figure 1, the impedances Z_1 and Z_2 are connected in parallel. Determine its equivalent impedance and represent it with Z_3 .

$$\begin{aligned} Z_3 &= Z_1 \parallel Z_2 \\ &= \frac{Z_1 Z_2}{Z_1 + Z_2} \\ &= \frac{(30 + j10)(20 - j50)}{(30 + j10) + (20 - j50)} \\ &= 26.09 - j5.122 \, \Omega \end{aligned}$$

Step 4 of 5

Redraw the circuit diagram by replacing the parallel combination of Z_1 and Z_2 with impedance Z_3 as shown in Figure 2.



Step 5 of 5

In Figure 2, the impedances Z_o and Z_3 are connected in series.

Calculate the input impedance, Z_{in} .

$$\begin{aligned} Z_{in} &= Z_o + Z_3 \\ &= 25 + j15 + 26.09 - j5.122 \\ &= (25 + 26.09) + j(15 - 5.122) \\ &= 51.09 + j9.878 \, \Omega \end{aligned}$$

Therefore, the input impedance, Z_{in} of the circuit is $51.09 + j9.878 \, \Omega$.